Kết quả bài tập thực hành 1 – DS200

Bài 1:

Bai1 > output_result > part-r-00000 > data

```
American Beauty (1999)  AverageRating: 3.75 (TotalRatings: 2)
Avatar (2009)   AverageRating: 4.75 (TotalRatings: 2)
Back to the Future (1985)   AverageRating: 4.00 (TotalRatings: 2)
Birdman (2014)  AverageRating: 3.00 (TotalRatings: 2)
Braveheart (1995)   AverageRating: 4.25 (TotalRatings: 2)
Coco (2017) AverageRating: 4.25 (TotalRatings: 2)
Dunkirk (2017)  AverageRating: 3.50 (TotalRatings: 2)
Fight Club (1999)   AverageRating: 4.25 (TotalRatings: 2)
Forrest Gump (1994) AverageRating: 3.75 (TotalRatings: 2)
Gladiator (2000)    AverageRating: 3.25 (TotalRatings: 2)
Good Will Hunting (1997)    AverageRating: 3.50 (TotalRatings: 2)
Gravity (2013)  AverageRating: 4.25 (TotalRatings: 2)
Inception (2010)    AverageRating: 3.75 (TotalRatings: 2)
Interstellar (2014) AverageRating: 3.50 (TotalRatings: 2)
Jurassic Park (1993)    AverageRating: 4.00 (TotalRatings: 2)
La La Land (2016)   AverageRating: 4.00 (TotalRatings: 2)
Mad Max: Fury Road (2015)   AverageRating: 4.00 (TotalRatings: 2)
Manchester by the Sea (2016)    AverageRating: 4.00 (TotalRatings: 2)
Memento (2000)  AverageRating: 4.00 (TotalRatings: 2)
Moonlight (2016)    AverageRating: 3.50 (TotalRatings: 2)
Parasite (2019) AverageRating: 4.00 (TotalRatings: 2)
Pulp Fiction (1994) AverageRating: 4.00 (TotalRatings: 2)
Saving Private Ryan (1998)  AverageRating: 4.00 (TotalRatings: 2)
Schindler's List (1993) AverageRating: 4.25 (TotalRatings: 2)
Se7en (1995)    AverageRating: 3.75 (TotalRatings: 2)
Slumdog Millionaire (2008)  AverageRating: 4.25 (TotalRatings: 2)
Spotlight (2015)    AverageRating: 4.25 (TotalRatings: 2)
Star Wars: A New Hope (1977)    AverageRating: 3.50 (TotalRatings: 2)
The Avengers (2012) AverageRating: 3.75 (TotalRatings: 2)
The Big Short (2015)    AverageRating: 4.50 (TotalRatings: 2)
The Dark Knight (2008)  AverageRating: 3.50 (TotalRatings: 2)
The Departed (2006) AverageRating: 3.25 (TotalRatings: 2)
The Godfather (1972)    AverageRating: 3.50 (TotalRatings: 2)
The Green Mile (1999)   AverageRating: 3.50 (TotalRatings: 2)
The Imitation Game (2014)   AverageRating: 3.75 (TotalRatings: 2)
The Lion King (1994)    AverageRating: 4.00 (TotalRatings: 2)
The Lord of the Rings: The Fellowship of the Ring (2001)    AverageRating: 4.25 (TotalRatings: 2)
The Lord of the Rings: The Return of the King (2003)    AverageRating: 3.50 (TotalRatings: 2)
The Lord of the Rings: The Two Towers (2002)    AverageRating: 3.75 (TotalRatings: 2)
The Matrix (1999)   AverageRating: 3.75 (TotalRatings: 2)
The Prestige (2006) AverageRating: 4.00 (TotalRatings: 2)
The Revenant (2015) AverageRating: 3.50 (TotalRatings: 2)
The Shawshank Redemption (1994) AverageRating: 4.00 (TotalRatings: 2)
The Silence of the Lambs (1991) AverageRating: 3.50 (TotalRatings: 2)
The Social Network (2010)   AverageRating: 3.75 (TotalRatings: 2)
The Usual Suspects (1995)   AverageRating: 4.25 (TotalRatings: 2)
The Wolf of Wall Street (2013)  AverageRating: 4.25 (TotalRatings: 2)
Titanic (1997)  AverageRating: 3.50 (TotalRatings: 2)
Toy Story (1995)    AverageRating: 4.25 (TotalRatings: 2)
Whiplash (2014) AverageRating: 3.75 (TotalRatings: 2)
```

Bài 2:

Bai2 > output_result > part-r-00000 > data

```
Action  Avg: 3.73, Count: 20
Adventure   Avg: 3.85, Count: 30
Animation   Avg: 4.17, Count: 6
Biography   Avg: 4.14, Count: 14
Children    Avg: 4.25, Count: 2
Comedy  Avg: 3.96, Count: 12
Crime   Avg: 3.78, Count: 20
Drama   Avg: 3.81, Count: 76
Family  Avg: 4.25, Count: 2
Fantasy Avg: 3.95, Count: 10
History Avg: 4.00, Count: 6
Music   Avg: 3.88, Count: 4
Mystery Avg: 4.00, Count: 8
Romance Avg: 3.75, Count: 8
Sci-Fi  Avg: 3.85, Count: 20
Thriller    Avg: 3.88, Count: 16
War Avg: 4.00, Count: 2
```

Bài 3:

Bai3 > output_result > part-r-00000 > data

```
American Beauty (1999)  Male: 3.00, Female: 4.50
Avatar (2009)   Male: 5.00, Female: 4.50
Back to the Future (1985)   Male: 4.00, Female: 4.00
Birdman (2014)  Male: 3.00, Female: 3.00
Braveheart (1995)   Male: 4.50, Female: 4.00
Coco (2017) Male: 5.00, Female: 3.50
Dunkirk (2017)  Male: 4.00, Female: 3.00
Fight Club (1999)   Male: 5.00, Female: 3.50
Forrest Gump (1994) Male: 2.50, Female: 5.00
Gladiator (2000)    Male: 3.50, Female: 3.00
Good Will Hunting (1997)    Male: 3.00, Female: 4.00
Gravity (2013)  Male: 4.00, Female: 4.50
Inception (2010)    Male: 3.50, Female: 4.00
Interstellar (2014) Male: 3.50, Female: 3.50
Jurassic Park (1993)    Male: 3.50, Female: 4.50
La La Land (2016)   Male: 5.00, Female: 3.00
Mad Max: Fury Road (2015)   Male: 4.00, Female: 4.00
Manchester by the Sea (2016)    Male: 3.50, Female: 4.50
Memento (2000)  Male: 3.50, Female: 4.50
Moonlight (2016)    Male: 3.00, Female: 4.00
Parasite (2019) Male: 4.00, Female: 4.00
Pulp Fiction (1994) Male: 4.50, Female: 3.50
Saving Private Ryan (1998)  Male: 4.00, Female: 4.00
Schindler's List (1993) Male: 4.00, Female: 4.50
Se7en (1995)    Male: 4.00, Female: 3.50
Slumdog Millionaire (2008)  Male: 4.50, Female: 4.00
Spotlight (2015)    Male: 4.50, Female: 4.00
Star Wars: A New Hope (1977)    Male: 4.00, Female: 3.00
The Avengers (2012) Male: 3.50, Female: 4.00
The Big Short (2015)    Male: 5.00, Female: 4.00
The Dark Knight (2008)  Male: 4.00, Female: 3.00
The Departed (2006) Male: 4.50, Female: 2.00
The Godfather (1972)    Male: 4.50, Female: 2.50
The Green Mile (1999)   Male: 3.00, Female: 4.00
The Imitation Game (2014)   Male: 4.50, Female: 3.00
The Lion King (1994)    Male: 4.50, Female: 3.50
The Lord of the Rings: The Fellowship of the Ring (2001)    Male: 4.50, Female: 4.00
The Lord of the Rings: The Return of the King (2003)    Male: 4.00, Female: 3.00
The Lord of the Rings: The Two Towers (2002)    Male: 4.50, Female: 3.00
The Matrix (1999)   Male: 3.00, Female: 4.50
The Prestige (2006) Male: 4.00, Female: 4.00
The Revenant (2015) Male: 3.50, Female: 3.50
The Shawshank Redemption (1994) Male: 3.00, Female: 5.00
The Silence of the Lambs (1991) Male: 3.50, Female: 3.50
The Social Network (2010)   Male: 3.50, Female: 4.00
The Usual Suspects (1995)   Male: 4.00, Female: 4.50
The Wolf of Wall Street (2013)  Male: 4.00, Female: 4.50
Titanic (1997)  Male: 4.00, Female: 3.00
Toy Story (1995)    Male: 4.50, Female: 4.00
Whiplash (2014) Male: 4.50, Female: 3.00
```

Bài 4:

Bai4 > output_result > part-r-00000 > data

```
American Beauty (1999)  [0-18: 0.00, 18-35: 3.75, 35-50: 0.00, 50+: 0.00]
Avatar (2009)   [0-18: 0.00, 18-35: 4.50, 35-50: 5.00, 50+: 0.00]
Back to the Future (1985)   [0-18: 0.00, 18-35: 4.00, 35-50: 0.00, 50+: 0.00]
Birdman (2014)  [0-18: 0.00, 18-35: 3.00, 35-50: 0.00, 50+: 0.00]
Braveheart (1995)   [0-18: 0.00, 18-35: 4.00, 35-50: 4.50, 50+: 0.00]
Coco (2017) [0-18: 0.00, 18-35: 4.25, 35-50: 0.00, 50+: 0.00]
Dunkirk (2017)  [0-18: 0.00, 18-35: 3.50, 35-50: 0.00, 50+: 0.00]
Fight Club (1999)   [0-18: 0.00, 18-35: 4.25, 35-50: 0.00, 50+: 0.00]
Forrest Gump (1994) [0-18: 0.00, 18-35: 3.75, 35-50: 0.00, 50+: 0.00]
Gladiator (2000)    [0-18: 0.00, 18-35: 3.00, 35-50: 3.50, 50+: 0.00]
Good Will Hunting (1997)    [0-18: 0.00, 18-35: 3.50, 35-50: 0.00, 50+: 0.00]
Gravity (2013)  [0-18: 0.00, 18-35: 4.50, 35-50: 4.00, 50+: 0.00]
Inception (2010)    [0-18: 0.00, 18-35: 3.75, 35-50: 0.00, 50+: 0.00]
Interstellar (2014) [0-18: 0.00, 18-35: 3.50, 35-50: 3.50, 50+: 0.00]
Jurassic Park (1993)    [0-18: 0.00, 18-35: 4.00, 35-50: 0.00, 50+: 0.00]
La La Land (2016)   [0-18: 0.00, 18-35: 3.00, 35-50: 5.00, 50+: 0.00]
Mad Max: Fury Road (2015)   [0-18: 0.00, 18-35: 4.00, 35-50: 0.00, 50+: 0.00]
Manchester by the Sea (2016)    [0-18: 0.00, 18-35: 4.50, 35-50: 3.50, 50+: 0.00]
Memento (2000)  [0-18: 0.00, 18-35: 4.50, 35-50: 3.50, 50+: 0.00]
Moonlight (2016)    [0-18: 0.00, 18-35: 4.00, 35-50: 3.00, 50+: 0.00]
Parasite (2019) [0-18: 0.00, 18-35: 4.00, 35-50: 4.00, 50+: 0.00]
Pulp Fiction (1994) [0-18: 0.00, 18-35: 4.00, 35-50: 0.00, 50+: 0.00]
Saving Private Ryan (1998)  [0-18: 0.00, 18-35: 4.00, 35-50: 4.00, 50+: 0.00]
Schindler's List (1993) [0-18: 0.00, 18-35: 4.50, 35-50: 4.00, 50+: 0.00]
Se7en (1995)    [0-18: 0.00, 18-35: 3.75, 35-50: 0.00, 50+: 0.00]
Slumdog Millionaire (2008)  [0-18: 0.00, 18-35: 4.00, 35-50: 4.50, 50+: 0.00]
Spotlight (2015)    [0-18: 0.00, 18-35: 4.00, 35-50: 4.50, 50+: 0.00]
Star Wars: A New Hope (1977)    [0-18: 0.00, 18-35: 3.00, 35-50: 4.00, 50+: 0.00]
The Avengers (2012) [0-18: 0.00, 18-35: 3.75, 35-50: 0.00, 50+: 0.00]
The Big Short (2015)    [0-18: 0.00, 18-35: 4.50, 35-50: 0.00, 50+: 0.00]
The Dark Knight (2008)  [0-18: 0.00, 18-35: 3.00, 35-50: 4.00, 50+: 0.00]
The Departed (2006) [0-18: 0.00, 18-35: 2.00, 35-50: 4.50, 50+: 0.00]
The Godfather (1972)    [0-18: 0.00, 18-35: 3.50, 35-50: 0.00, 50+: 0.00]
The Green Mile (1999)   [0-18: 0.00, 18-35: 4.00, 35-50: 3.00, 50+: 0.00]
The Imitation Game (2014)   [0-18: 0.00, 18-35: 3.00, 35-50: 4.50, 50+: 0.00]
The Lion King (1994)    [0-18: 0.00, 18-35: 4.00, 35-50: 0.00, 50+: 0.00]
The Lord of the Rings: The Fellowship of the Ring (2001)    [0-18: 0.00, 18-35: 4.00, 35-50: 4.50, 50+: 0.00]
The Lord of the Rings: The Return of the King (2003)    [0-18: 0.00, 18-35: 3.50, 35-50: 0.00, 50+: 0.00]
The Lord of the Rings: The Two Towers (2002)    [0-18: 0.00, 18-35: 3.00, 35-50: 4.50, 50+: 0.00]
The Matrix (1999)   [0-18: 0.00, 18-35: 3.75, 35-50: 0.00, 50+: 0.00]
The Prestige (2006) [0-18: 0.00, 18-35: 4.00, 35-50: 4.00, 50+: 0.00]
The Revenant (2015) [0-18: 0.00, 18-35: 3.50, 35-50: 0.00, 50+: 0.00]
The Shawshank Redemption (1994) [0-18: 0.00, 18-35: 4.00, 35-50: 0.00, 50+: 0.00]
The Silence of the Lambs (1991) [0-18: 0.00, 18-35: 3.50, 35-50: 3.50, 50+: 0.00]
The Social Network (2010)   [0-18: 0.00, 18-35: 4.00, 35-50: 3.50, 50+: 0.00]
The Usual Suspects (1995)   [0-18: 0.00, 18-35: 4.25, 35-50: 0.00, 50+: 0.00]
The Wolf of Wall Street (2013)  [0-18: 0.00, 18-35: 4.50, 35-50: 4.00, 50+: 0.00]
Titanic (1997)  [0-18: 0.00, 18-35: 3.00, 35-50: 4.00, 50+: 0.00]
Toy Story (1995)    [0-18: 0.00, 18-35: 4.00, 35-50: 4.50, 50+: 0.00]
Whiplash (2014) [0-18: 0.00, 18-35: 3.00, 35-50: 4.50, 50+: 0.00]
```